

$$\begin{aligned}
& \frac{1}{16\pi^2} \left(-\frac{1}{288} \frac{1}{m_\phi^2} \left(s_\gamma^2 \overline{y_d}^{i4p} \left(y_d^{i3p} \left(18 c_\gamma^2 \overline{y_u}^{i2r} y_u^{i1r} + g_2^2 \delta_{i1i2} \right) + 4 g_3^2 y_d^{i1p} \delta_{i2i3} \right) + s_\gamma^2 \right. \right. \\
& \quad \overline{y_d}^{i2p} \left(4 g_3^2 y_d^{i3p} \delta_{i1i4} + y_d^{i1p} \left(18 c_\gamma^2 \left(\overline{y_d}^{i4r} y_d^{i3r} + \overline{y_u}^{i4r} y_u^{i3r} \right) + g_2^2 \delta_{i3i4} \right) \right) + \\
& \quad c_\gamma^2 \left(\overline{y_u}^{i4p} \left(g_2^2 y_u^{i3p} \delta_{i1i2} + 4 g_3^2 y_u^{i1p} \delta_{i2i3} \right) + \right. \\
& \quad \left. \left. \overline{y_u}^{i2p} \left(4 g_3^2 y_u^{i3p} \delta_{i1i4} + y_u^{i1p} \left(18 s_\gamma^2 \overline{y_u}^{i4r} y_u^{i3r} + g_2^2 \delta_{i3i4} \right) \right) \right) \right) + \\
& \quad \frac{1}{18} g_2^4 \text{LF}_{3,0}[\mathbf{m}_2] \delta_{i1i2} \delta_{i3i4} + \frac{1}{12} g_2^4 \text{LF}_{4,-1}[\mathbf{m}_2] \delta_{i1i2} \delta_{i3i4} - \frac{4}{45} g_2^4 \text{LF}_{5,-2}[\mathbf{m}_2] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{1}{12} g_3^4 \text{LF}_{3,0}[\mathbf{m}_3] \delta_{i1i4} \delta_{i2i3} + \frac{1}{8} g_3^4 \text{LF}_{4,-1}[\mathbf{m}_3] \delta_{i1i4} \delta_{i2i3} - \\
& \quad \frac{2}{15} g_3^4 \text{LF}_{5,-2}[\mathbf{m}_3] \delta_{i1i4} \delta_{i2i3} + \frac{1}{36} \sum_{\mathbf{p}} g_3^4 \text{LF}_{3,0}[\mathbf{m}_d^{\mathbf{p}}] \delta_{i1i4} \delta_{i2i3} - \\
& \quad \frac{5}{96} \sum_{\mathbf{p}} g_3^4 \text{LF}_{4,-1}[\mathbf{m}_d^{\mathbf{p}}] \delta_{i1i4} \delta_{i2i3} + \\
& \quad \frac{1}{45} \sum_{\mathbf{p}} g_3^4 \text{LF}_{5,-2}[\mathbf{m}_d^{\mathbf{p}}] \delta_{i1i4} \delta_{i2i3} + \frac{1}{36} \sum_{\mathbf{p}} g_2^4 \text{LF}_{3,0}[\mathbf{m}_l^{\mathbf{p}}] \delta_{i1i2} \delta_{i3i4} - \\
& \quad \frac{5}{96} \sum_{\mathbf{p}} g_2^4 \text{LF}_{4,-1}[\mathbf{m}_l^{\mathbf{p}}] \delta_{i1i2} \delta_{i3i4} + \frac{1}{45} \sum_{\mathbf{p}} g_2^4 \text{LF}_{5,-2}[\mathbf{m}_l^{\mathbf{p}}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{1}{36} \sum_{\mathbf{p}} \left(2 g_3^4 \delta_{i1i4} \delta_{i2i3} + 3 g_2^4 \delta_{i1i2} \delta_{i3i4} \right) \text{LF}_{3,0}[\mathbf{m}_q^{\mathbf{p}}] - \\
& \quad \frac{5}{96} \sum_{\mathbf{p}} \left(2 g_3^4 \delta_{i1i4} \delta_{i2i3} + 3 g_2^4 \delta_{i1i2} \delta_{i3i4} \right) \text{LF}_{4,-1}[\mathbf{m}_q^{\mathbf{p}}] + \\
& \quad \frac{1}{45} \sum_{\mathbf{p}} \left(2 g_3^4 \delta_{i1i4} \delta_{i2i3} + 3 g_2^4 \delta_{i1i2} \delta_{i3i4} \right) \text{LF}_{5,-2}[\mathbf{m}_q^{\mathbf{p}}] + \frac{1}{36} \sum_{\mathbf{p}} g_3^4 \text{LF}_{3,0}[\mathbf{m}_u^{\mathbf{p}}] \delta_{i1i4} \delta_{i2i3} - \\
& \quad \frac{5}{96} \sum_{\mathbf{p}} g_3^4 \text{LF}_{4,-1}[\mathbf{m}_u^{\mathbf{p}}] \delta_{i1i4} \delta_{i2i3} + \frac{1}{45} \sum_{\mathbf{p}} g_3^4 \text{LF}_{5,-2}[\mathbf{m}_u^{\mathbf{p}}] \delta_{i1i4} \delta_{i2i3} + \\
& \quad \frac{1}{24} \left(s_\gamma^2 \overline{y_d}^{i2p} \left(3 c_\gamma^2 y_d^{i1p} \left(\overline{y_d}^{i4r} y_d^{i3r} - \overline{y_u}^{i4r} y_u^{i3r} \right) - g_3^2 y_d^{i3p} \delta_{i1i4} \right) - \right. \\
& \quad s_\gamma^2 \overline{y_d}^{i4p} \left(3 c_\gamma^2 y_d^{i3p} \overline{y_u}^{i2r} y_u^{i1r} + g_3^2 y_d^{i1p} \delta_{i2i3} \right) + \\
& \quad c_\gamma^2 \left(\overline{y_u}^{i2p} \left(3 s_\gamma^2 \overline{y_u}^{i4r} y_u^{i1p} y_u^{i3r} - g_3^2 y_u^{i3p} \delta_{i1i4} \right) - g_3^2 \overline{y_u}^{i4p} y_u^{i1p} \delta_{i2i3} \right) \left. \right) \\
& \quad \text{LF}_{1,2}[\mathbf{m}_\oplus] + \frac{1}{48} \left(s_\gamma^2 \overline{y_d}^{i4p} y_d^{i3p} \left(3 c_\gamma^2 \overline{y_u}^{i2r} y_u^{i1r} - g_2^2 \delta_{i1i2} \right) + \right. \\
& \quad s_\gamma^2 \overline{y_d}^{i2p} y_d^{i1p} \left(3 s_\gamma^2 \overline{y_d}^{i4r} y_d^{i3r} + 3 c_\gamma^2 \overline{y_u}^{i4r} y_u^{i3r} - g_2^2 \delta_{i3i4} \right) + \\
& \quad c_\gamma^2 \left(-g_2^2 \overline{y_u}^{i4p} y_u^{i3p} \delta_{i1i2} + \overline{y_u}^{i2p} y_u^{i1p} \left(3 c_\gamma^2 \overline{y_u}^{i4r} y_u^{i3r} - g_2^2 \delta_{i3i4} \right) \right) \left. \right) \text{LF}_{2,1}[\mathbf{m}_\oplus] + \\
& \quad \frac{1}{144} g_2^2 \left(3 \left(s_\gamma^2 \overline{y_d}^{i4p} y_d^{i3p} + c_\gamma^2 \overline{y_u}^{i4p} y_u^{i3p} \right) \delta_{i1i2} + \right. \\
& \quad \left. \left(3 s_\gamma^2 \overline{y_d}^{i2p} y_d^{i1p} + 3 c_\gamma^2 \overline{y_u}^{i2p} y_u^{i1p} + 4 g_2^2 \delta_{i1i2} \right) \delta_{i3i4} \right) \text{LF}_{3,0}[\mathbf{m}_\oplus] - \\
& \quad \frac{5}{96} g_2^4 \text{LF}_{4,-1}[\mathbf{m}_\oplus] \delta_{i1i2} \delta_{i3i4} + \frac{1}{45} g_2^4 \text{LF}_{5,-2}[\mathbf{m}_\oplus] \delta_{i1i2} \delta_{i3i4} + \frac{1}{36} g_2^4 \text{LF}_{3,0}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{1}{24} g_2^4 \text{LF}_{4,-1}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} - \frac{2}{45} g_2^4 \text{LF}_{5,-2}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} + \\
& \quad \frac{1}{1728} g_1^2 \left(g_3^2 \delta_{i1i4} \delta_{i2i3} + g_2^2 \delta_{i1i2} \delta_{i3i4} \right) \text{LF}_{2,1,0}[\mathbf{m}_1, \mathbf{m}_q^{i1}] + \\
& \quad \frac{1}{1728} g_1^2 \left(g_3^2 \delta_{i1i4} \delta_{i2i3} + g_2^2 \delta_{i1i2} \delta_{i3i4} \right) \text{LF}_{2,2,-1}[\mathbf{m}_1, \mathbf{m}_q^{i1}] - \\
& \quad \frac{1}{864} g_1^2 \left(g_3^2 \delta_{i1i4} \delta_{i2i3} + g_2^2 \delta_{i1i2} \delta_{i3i4} \right) \text{LF}_{3,1,-1}[\mathbf{m}_1, \mathbf{m}_q^{i1}] + \\
& \quad \frac{1}{1728} g_1^2 \left(g_3^2 \delta_{i1i4} \delta_{i2i3} + g_2^2 \delta_{i1i2} \delta_{i3i4} \right) \text{LF}_{4,1,-2}[\mathbf{m}_1, \mathbf{m}_q^{i1}] + \\
& \quad \frac{1}{1728} g_1^2 \left(g_3^2 \delta_{i1i4} \delta_{i2i3} + g_2^2 \delta_{i1i2} \delta_{i3i4} \right) \text{LF}_{2,1,0}[\mathbf{m}_1, \mathbf{m}_q^{i2}] + \\
& \quad \frac{1}{1728} g_1^2 \left(g_3^2 \delta_{i1i4} \delta_{i2i3} + g_2^2 \delta_{i1i2} \delta_{i3i4} \right) \text{LF}_{2,2,-1}[\mathbf{m}_1, \mathbf{m}_q^{i2}] - \\
& \quad \frac{1}{864} g_1^2 \left(g_3^2 \delta_{i1i4} \delta_{i2i3} + g_2^2 \delta_{i1i2} \delta_{i3i4} \right) \text{LF}_{3,1,-1}[\mathbf{m}_1, \mathbf{m}_q^{i2}] + \\
& \quad \frac{1}{1728} g_1^2 \left(g_3^2 \delta_{i1i4} \delta_{i2i3} + g_2^2 \delta_{i1i2} \delta_{i3i4} \right) \text{LF}_{4,1,-2}[\mathbf{m}_1, \mathbf{m}_q^{i2}] + \\
& \quad \frac{1}{1728} g_1^2 \left(g_3^2 \delta_{i1i4} \delta_{i2i3} + g_2^2 \delta_{i1i2} \delta_{i3i4} \right) \text{LF}_{2,1,0}[\mathbf{m}_1, \mathbf{m}_q^{i3}] + \\
& \quad \frac{1}{1728} g_1^2 \left(g_3^2 \delta_{i1i4} \delta_{i2i3} + g_2^2 \delta_{i1i2} \delta_{i3i4} \right) \text{LF}_{2,2,-1}[\mathbf{m}_1, \mathbf{m}_q^{i3}] - \\
& \quad \frac{1}{864} g_1^2 \left(g_3^2 \delta_{i1i4} \delta_{i2i3} + g_2^2 \delta_{i1i2} \delta_{i3i4} \right) \text{LF}_{3,1,-1}[\mathbf{m}_1, \mathbf{m}_q^{i3}] + \\
& \quad \frac{1}{1728} g_1^2 \left(g_3^2 \delta_{i1i4} \delta_{i2i3} + g_2^2 \delta_{i1i2} \delta_{i3i4} \right) \text{LF}_{4,1,-2}[\mathbf{m}_1, \mathbf{m}_q^{i3}] + \\
& \quad \frac{1}{1728} g_1^2 \left(g_3^2 \delta_{i1i4} \delta_{i2i3} + g_2^2 \delta_{i1i2} \delta_{$$